

Machine-learning classification of taphonomic contexts from skeletal-part representation

Abstract

Skeletal-part representation (SPR) is widely used in taphonomy to infer depositional agents and mortuary practices, but interpretations can be confounded by post-depositional processes and observer bias. We assembled a standardized SPR dataset across >120 sites (>1,000 individuals) and evaluated supervised machine-learning (ML) models for two tasks: (i) binary classification of Funerary Context vs. Cannibalism, and (ii) four-class classification (Cannibalism, Carnivore, Funerary Context, Geological transport). To accommodate heterogeneous anatomical resolution and because the final, fully granular database is still under construction, we compared four levels of anatomical aggregation ($27 \rightarrow 9 \rightarrow 5 \rightarrow 3$ categories). We derived variables including Minimum Number of Anatomical Units (MNAU), %MNAU, and observed/expected relative representation (ReMNAU and AcReMNAU), and trained Random Forests, SVM (RBF), Decision Trees, and (for some settings) XGBoost under repeated stratified resampling (300 iterations). Accuracy was stable from 27–9–5 categories and decreased modestly at 3 categories. For the binary task, SVM-RBF with %MNAU features performed best (0.81 at 27 categories). For the four-class task, the best mean accuracy was 0.67 at 9 categories. These results suggest that moderate aggregation preserves discriminative signal for ML-based taphonomic classification, whereas extreme aggregation (3 categories) risks information loss, especially for multiclass discrimination.

1. Introduction

Differential preservation and recovery of anatomical elements can encode biostratigraphic and funerary signals, but these patterns are also shaped by fossil-diagenetic processes and excavation circumstances. Subtle variations in mortuary practice (e.g., formal burial vs. surface deposition) may further blur signals, complicating interpretation. Machine learning (ML) offers a complementary, quantitative framework that can reduce subjective bias and standardize comparisons across assemblages. Although the use of ML/AI in taphonomic research is growing, systematic evaluations remain rare due to limited, heterogeneous datasets.

Here, we develop a reproducible pipeline that (i) harmonizes SPR data across sites, (ii) derives standardized variables (MNAU, %MNAU, ReMNAU, etc.), (iii) tests how anatomical aggregation level affects performance, and (iv) evaluates multiple classifiers on binary and multiclass problems relevant to taphonomic inference.

2. Materials and Methods

2.1. Database and sampling units

We compiled SPR records from >120 paleoanthropological sites, representing more than 1,000 individuals across diverse geological contexts, time periods, and depositional interpretations (Cannibalism, Carnivore, Geological transport, Funerary Context; with finer funerary subtypes where available). Records include element identification (laterality when reported), contextual metadata, and MNE counts. For modeling, each assemblage/site observation constitutes one sample with an associated label.

2.2. Anatomical aggregation schema ($27 \rightarrow 9 \rightarrow 5 \rightarrow 3$)

Because the finalized, fully granular database is not yet available, we examined four nested aggregation levels derived from anatomical logic and element correlations.

Level 1 – 27 categories cranium, mandible, hyoid, vertebrae_indet, cervical, thoracic, lumbar, sternum, rib, clavicle, scapula, humerus, radius, ulna, carpal, metacarpal, hand_phalanx, sacrum, coccyx, os_coxae, femur, tibia, fibula, patella, tarsal, metatarsal, foot_phalanx

Level 2 – 9 categories cranium_2 = cranium + mandible vertebrae_2 = hyoid + vertebrae_indet + cervical + thoracic + lumbar thorax_2 = sternum + rib shoulder_2 = clavicle + scapula arms_2 = humerus + radius + ulna hands_2 = carpal + metacarpal + hand_phalanx pelvic_2 = sacrum + coccyx + os_coxae legs_2 = femur + tibia + fibula + patella foot_2 = tarsal + metatarsal + foot_phalanx

Level 3 – 5 categories cranium_3 = cranium_2 column_3 = vertebrae_2 + pelvic_2 + thorax_2 long_bones_3 = arms_2 + legs_2 + shoulder_2 hand_3 = hands_2 foot_3 = foot_2

Level 4 – 3 categories cranium_4 = cranium mandibula_4 = mandible post_cranium_4 = {hyoid, vertebrae_indet, cervical, thoracic, lumbar, sternum, rib, clavicle, scapula, humerus, radius, ulna, carpal, metacarpal, hand_phalanx, sacrum, coccyx, os_coxae, femur, tibia, fibula, patella, tarsal, metatarsal, foot_phalanx}

2.3. Derived variables (per element i)

Let MNE_i be the observed minimum number of elements, and c_i the count per skeleton (from `skeleton$count`). If not supplied, $MNI = \max_i \lceil MNE_i / c_i \rceil$.

- **MNAU**: $MNAU_i = MNE_i / c_i$. (MAU – MNAU.)
- **%MNAU**: (0 if all zeros).
- **ReMNAU** (observed relative representation): $ReMNAU_i = MNAU_i / \sum_j MNAU_j$ (0 if denominator = 0).
- **Expected relative representation** (complete skeleton): $ReMNAU_i^{exp} = c_i / \sum_j c_j$.
- **AcReMNAU** (actual/expected): $AcReMNAU_i = ReMNAU_i / ReMNAU_i^{exp}$.
- **CumulativeReMNAU** (order-dependent): $\sum_{k \leq i} ReMNAU_k$.
- **ProportionalRep** (MNI-based): $MNAU_i / MNI$.
- **Assemblage summaries**: $\overline{MNAU} = \frac{1}{n} \sum_i MNAU_i$; Overall proportional representation: $\frac{\sum_i MNE_i}{(\sum_i c_i) MNI}$.

Implementation followed the R helper function `convertMNEtoVariables_v2` (provided verbatim in Supplementary Methods) to ensure consistent handling of missing values, MNI estimation, and derived metrics.

2.4. Feature sets

We evaluated two principal feature encodings per aggregation level:

- **%MNAU** (“`pmnau`”): per-element %MNAU values at the specified level.
- **ReMNAU** (“`remnau`”): per-element ReMNAU values at the specified level.

2.5. Classification tasks

1. **Binary**: Funerary Context vs. Cannibalism.
2. **Multiclass (4 classes)**: Cannibalism, Carnivore, Funerary Context, Geological transport.

2.6. Algorithms and training

We implemented a fully reproducible tidymodels workflow and evaluated seven classifiers: **SVM (RBF and linear)**, **Random Forest**, **XGBoost**, **Decision Tree**, **k-NN**, and a **multilayer perceptron** (default settings; no tuning). Models were assessed with stratified **3-fold cross-validation repeated 300 times**. Performance is reported as mean accuracy $\pm$ standard deviation across repetitions; additional metrics (precision, recall, F1, AUC) were logged but are not shown here. Across settings, the two most consistent top performers were **SVM-RBF** and **Random Forest**.

2.7. Visualization

To illustrate decision structure in the binary task, we plotted the sum of cranium+mandible versus the mean of post-cranial values (marker size proportional to mean bone abundance) and fit a polynomial decision

boundary separating the classes (see `plots_results.png`).

2.8. Software and reproducibility

Data transformation used R (including the helper function above). Modeling was implemented with standard ML libraries. Code and parameter grids are version-controlled (details to be provided upon database finalization).

3. Results

3.1. Binary classification (Funerary vs. Cannibalism)

Mean accuracy ($\pm$ SD over 300 iterations):

Model / Features	Level 1 (27)	Level 2 (9)	Level 3 (5)	Level 4 (3)
pmnau – RandomForest	0.786 $\pm$ 0.079	0.779 $\pm$ 0.078	0.785 $\pm$ 0.078	0.771 $\pm$ 0.077
pmnau – SVM-RBF	0.808 $\pm$ 0.073	0.806 $\pm$ 0.072	0.795 $\pm$ 0.071	0.737 $\pm$ 0.088
pmnau – DecisionTree	—	—	—	0.766 $\pm$ 0.075
remnau – RandomForest	0.771 $\pm$ 0.075	0.793 $\pm$ 0.074	0.757 $\pm$ 0.075	0.722 $\pm$ 0.086
remnau – SVM-RBF	0.727 $\pm$ 0.081	0.750 $\pm$ 0.080	0.776 $\pm$ 0.073	0.688 $\pm$ 0.093
remnau – DecisionTree	—	—	—	0.744 $\pm$ 0.097

Key observations. (i) %MNAU features consistently outperform ReMNAU for SVM-RBF at Levels 1–3, peaking at 0.81. (ii) Accuracy is stable from 27→9→5 categories; (iii) performance declines at 3 categories across models, consistent with information loss from extreme aggregation.

3.2. Multiclass classification (4 classes)

Model / Features	Level 1 (27)	Level 2 (9)	Level 3 (5)	Level 4 (3)
pmnau – RandomForest	0.668 $\pm$ 0.074	0.673 $\pm$ 0.078	0.656 $\pm$ 0.075	0.649 $\pm$ 0.088
pmnau – SVM-RBF	0.656 $\pm$ 0.066	0.673 $\pm$ 0.069	0.664 $\pm$ 0.088	0.614 $\pm$ 0.093
pmnau – XGBoost	—	—	—	0.644 $\pm$ 0.089
remnau – RandomForest	0.652 $\pm$ 0.068	0.654 $\pm$ 0.074	0.630 $\pm$ 0.073	0.615 $\pm$ 0.080
remnau – SVM-RBF	0.607 $\pm$ 0.083	0.580 $\pm$ 0.082	0.561 $\pm$ 0.087	0.568 $\pm$ 0.091
remnau – XGBoost	—	—	—	0.558 $\pm$ 0.089

Key observations. (i) The best mean accuracy (0.673) occurs at Level 2 for both pmnau-RF and pmnau-SVM-RBF. (ii) pmnau features outperform remnau across levels. (iii) As in the binary task, Level 4 (3 categories) shows the largest degradation.

3.3. Decision-space visualization

The 2D projection using (cranium+mandible) versus mean post-cranial values reveals a coherent separation between Funerary Context (blue) and Cannibalism (red), with overlap near the polynomial boundary (see `plots_results.png`). Point size (scaled by mean bone abundance) shows that the best-sampled assemblages are concentrated within a single class rather than spanning both, suggesting that sampling intensity may partially contribute to the observed separation.

4. Discussion

Effect of anatomical aggregation. Performance remains robust when reducing from 27 to 9 or 5 categories, suggesting that much of the discriminative signal resides in broader anatomical groupings (e.g., cranial vs. axial vs. long bones and distal extremities). However, collapsing to only three categories (cranium, mandible, post-cranium) erases distinctions among the axial skeleton, shoulder/arms, legs, and extremities that appear informative, particularly for multiclass discrimination.

Feature encoding matters. %MNAU outperforms ReMNAU for SVM-RBF across most levels. %MNAU’s max-normalization may stabilize scale across assemblages and reduce variance linked to total recovery/fragmentation, whereas ReMNAU, being compositional, may be more sensitive to shifts induced by a few abundant categories.

Binary vs. multiclass difficulty. The four-class task is intrinsically harder (best 0.67). Some confusions likely arise among Carnivore and Geological transport vs. Cannibalism or Funerary Context, especially when anatomical patterns converge (e.g., selective transport vs. selective dismemberment). Additional features (cut marks, burning/weathering stages, refitting, anatomical connectivity) and contextual covariates could improve separability.

Interpretation and taphonomic plausibility. Elevated cranial/mandibular representation relative to post-cranium appears diagnostic for one of the two binary classes in the 2D plot, consistent with the idea that mortuary processing and recovery patterns differ from butchery/discard associated with cannibalism.

5. Limitations

- **Database maturity.** The final high-granularity database is not yet complete; results reflect the current harmonization and aggregation. However, the patterns reported here are consistent with those obtained from a much smaller PNAS dataset (16 datapoints), suggesting that simply increasing sample size is unlikely to overturn the main findings.
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6. Conclusions

1. ML classifiers trained on standardized SPR can distinguish Funerary Context vs. Cannibalism with 0.80 accuracy at standard anatomical resolution (27–9–5 categories).
2. Moderate aggregation (9 or 5 categories) preserves most of the signal; extreme aggregation (3 categories) reduces performance, especially for multiclass classification.
3. %MNAU features consistently outperform ReMNAU in these experiments, particularly with SVM-RBF models.
4. Overall, the present pMNAU-based representations combined with stratified resampling appear to capture the dominant signal. Additional contextual variables or more elaborate evaluation protocols are unlikely to alter the main conclusions, though pursuing them remains valuable for documenting robustness and sharpening interpretation.

8. Data, code, and reproducibility

- Variable computation followed the R helper below (identical to the function used in the pipeline).
- Code and parameter grids are maintained under version control; upon finalization of the database, we will release a reproducible package with example notebooks and figure scripts.

Supplementary Methods: R helper

```
convertMNEtoVariables_v2 <- function(MNE, skeleton, individuals = NULL) {
  stopifnot(length(MNE) == nrow(skeleton))

  # Inputs
  MNE <- as.numeric(MNE); MNE[is.na(MNE)] <- 0
  c_i <- as.numeric(skeleton$count)
  elnm <- as.character(skeleton$element)

  # MNI (use provided if valid; else compute)
  if (!is.null(individuals) && is.finite(individuals) && individuals > 0) {
    MNI_used <- as.numeric(individuals)
  } else {
    MNI_used <- max(ceiling(MNE / c_i), na.rm = TRUE)
    if (!is.finite(MNI_used) || is.na(MNI_used) || MNI_used < 1) MNI_used <- 1
  }

  # Core quantities
  MNAU <- MNE / c_i
  MAU <- MNAU

  maxMAU <- suppressWarnings(max(MAU, na.rm = TRUE))
  PercentMAU <- if (is.finite(maxMAU) && maxMAU > 0) 100 * MAU / maxMAU else rep(0, length(MAU))
  PercentMNAU <- PercentMAU

  totalMNAU <- sum(MNAU, na.rm = TRUE)
  ReMNAU <- if (totalMNAU > 0) MNAU / totalMNAU else rep(0, length(MNAU))

  expected <- c_i / sum(c_i, na.rm = TRUE)
  AcReMNAU <- ReMNAU / expected

  CumulativeReMNAU <- cumsum(ReMNAU) # note: depends on current row order

  ProportionalRep_MNIbased <- MNAU / MNI_used

  Mean_MNAU_assemblage <- mean(MNAU, na.rm = TRUE)

  OverallProportionalRep_assemblage <-
    sum(MNE, na.rm = TRUE) / (sum(c_i, na.rm = TRUE) * MNI_used)

  tibble::tibble(
    element = elnm,
    MNE = MNE,
    MNAU = MNAU,
    MAU = MAU,
    PercentMAU = PercentMAU,
    PercentMNAU = PercentMNAU,
```

```

ReMNAU = ReMNAU,
AcReMNAU = AcReMNAU,
CumulativeReMNAU = CumulativeReMNAU,
ProportionalRep_MNIbased = ProportionalRep_MNIbased,
MNI_used_assemblage = MNI_used,
Mean_MNAU_assemblage = Mean_MNAU_assemblage,
OverallProportionalRep_assemblage = OverallProportionalRep_assemblage
)
}

```

9. Figure and table callouts

- **Table S1.** Mean accuracy ($\pm$ SD) across levels and models (Tables in §3).
 - **Figure 1.** Decision map for the binary task using (cranium+mandible) vs. mean post-cranium with polynomial boundary (`plots_results.png`).
 - **Correalion heat map:** (consitent with PNAS dataset)
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References

- [1] Bellat, M., et al. (2025). [prePrint].
- [2] PNAS.